

1. A bank wants to predict whether a loan applicant will default based on credit score, income, and past loan history. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Classification (because the output is a category: whether a loan applicant will default or not).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the data into training and testing sets.
 5. Choose a classification model and train it.
 6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
 7. Fine-tune using GridSearchCV and deploy the best model
2. A retail store wants to predict the demand for different products to optimize inventory levels. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Regression (because the output is a continuous value: predict the demand for different products).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
 5. Choose a regression model and train the model using the training data.
 6. Evaluate model performance using metrics like R^2 score on the test data.
 7. Fine-tune the model with GridSearch CV and then deploy the best model
3. A factory wants to detect whether a manufactured product is defective based on sensor readings and quality control data. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Classification (because the output is a category: whether the product is defective or not).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model

4. A healthcare provider wants to analyze patient symptoms and classify them into different disease categories. What type of ML problem is this, and what steps would you take to solve it?

A: Unsupervised Learning – Clustering (because there are no labels, and the goal is to group patient symptoms into different disease categories).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Select relevant features and scale the features using StandardScaler.
 3. Choose a clustering algorithm.
 4. Determine the optimal number of clusters using methods like the Elbow Method or Silhouette Score.
 5. Train the model and assign cluster labels to customers
 6. Analyze each cluster and use insights to decide disease categories.
5. An e-commerce company wants to identify and remove fake reviews posted by bots or fraudsters. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Classification (because the output is a category: whether it is fake or genuine review).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
 2. Encode categorical using OneHotEncoder or LabelEncoder.
 3. Scale numerical features using StandardScaler.
 4. Split the data into training and testing sets.
 5. Choose a classification model and train it.
 6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
 7. Fine-tune using GridSearchCV and deploy the best model to find and remove fake review
6. A financial firm wants to predict stock price movements based on historical price data and market indicators. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Regression (because the output is a continuous value: predict stock price).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the dataset into training and testing sets (e.g., 80% train, 20% test).
5. Choose a regression model and train the model using the training data.
6. Evaluate model performance using metrics like R^2 score on the test data.
7. Fine-tune the model with GridSearch CV and then deploy the best model

7. A social media platform wants to detect fake user accounts based on user activity and profile data. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Classification (because the output is a category: whether it is fake or genuine user accounts).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model to find fake user accounts

8. A marketing agency wants to segment customers into different groups based on their purchasing behavior. What type of ML problem is this, and what steps would you take to solve it?

A: Unsupervised Learning – Clustering (because there are no labels, and the goal is to group customers).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Select relevant features and scale the features using StandardScaler.
3. Choose a clustering algorithm.
4. Determine the optimal number of clusters using methods like the Elbow Method or Silhouette Score.
5. Train the model and assign cluster labels to customers
6. Analyze each cluster and use insights to decide customers categories.

9. A geospatial research team wants to analyze satellite images to classify different land types (forest, water, urban). What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Classification (because the output is a category: different land types).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model to find different land types

10. A streaming service wants to predict which users are likely to cancel their subscriptions. What type of ML problem is this, and what steps would you take to solve it?

A: Supervised Learning – Classification (because the output is a category: users are likely to keep or cancel their subscriptions).

Step by Step Logic:

1. Use data given, Clean the data by handling missing values and outliers.
2. Encode categorical using OneHotEncoder or LabelEncoder.
3. Scale numerical features using StandardScaler.
4. Split the data into training and testing sets.
5. Choose a classification model and train it.
6. Evaluate performance using metrics like ROC-AUC score and F1 Weighted Score
7. Fine-tune using GridSearchCV and deploy the best model to find users are likely to cancel their subscriptions